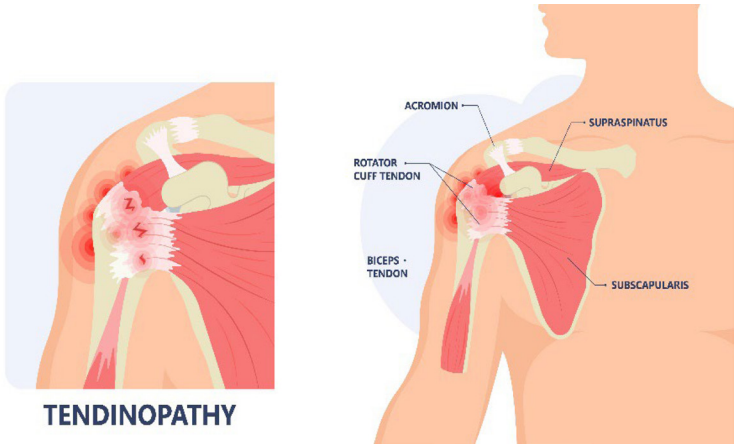
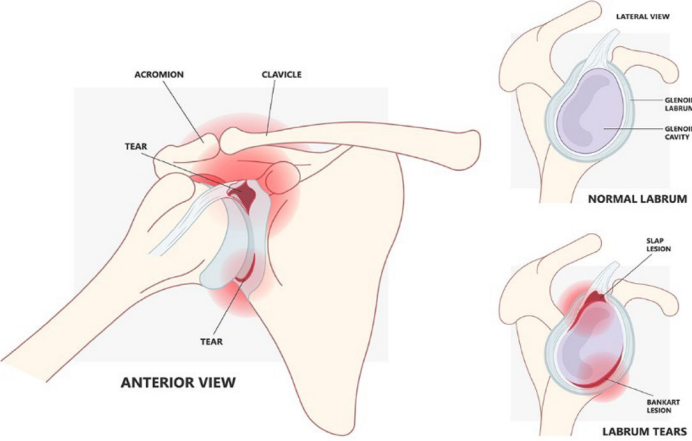


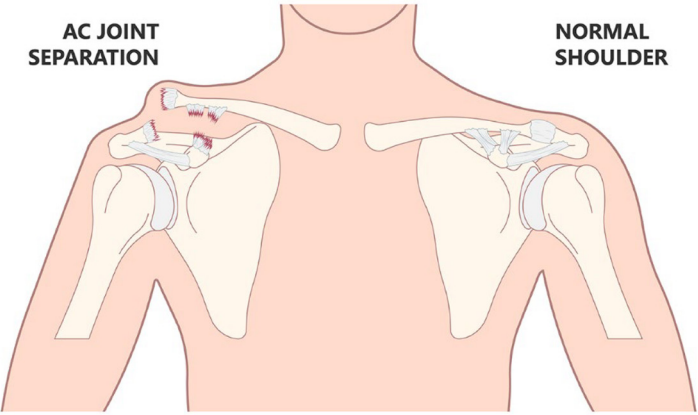
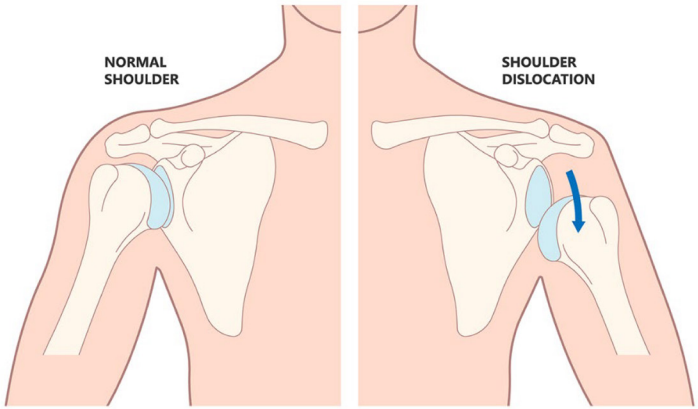
Common Shoulder Injuries in Bodybuilding



Injury Type	General Information	Symptoms	Relevant Risk Factors
Rotator Cuff Tendinopathy  TENDINOPATHY <i>Adapted from Pemporon/Shutterstock.com</i>	Irritation of one (or more) of the tendons of one of four rotator cuff muscles (supraspinatus, infraspinatus, teres minor, or subscapularis); most commonly supraspinatus; can be either insidious or traumatic (fall on an outstretched hand), but typically occurs insidiously due to overuse.	Pain with shoulder range of motion, particularly mid-range and overhead; possible anterior (front) shoulder pain near the biceps brachii; painful/weak external rotation and/or abduction (infraspinatus/teres minor/supraspinatus) or internal rotation (subscapularis) that increases with load; possible shoulder impingement.	Overhead movements and chronic use of upper extremity exercises without adequate rest; shoulder external rotation weakness.
Labral Tear (SLAP)  LABRUM TEARS <i>Adapted from Pemporon/Shutterstock.com</i>	Superior labral tears anterior to posterior (SLAP) can be a cause of shoulder pain; tear occurs when the top portion of the connective tissue surrounding the shoulder is torn with or without biceps brachii involvement and normally occurs over time and is not traumatic.	Reports of clicking, pain, instability, or locking with relevant functional impairments, particularly overhead.	Scapular dyskinesis (excessively anteriorly tipped, internally rotated scapula) increases tensile stress on the biceps brachii during movement; loss in shoulder rotational range of motion; overhead activities, most typically throwing.

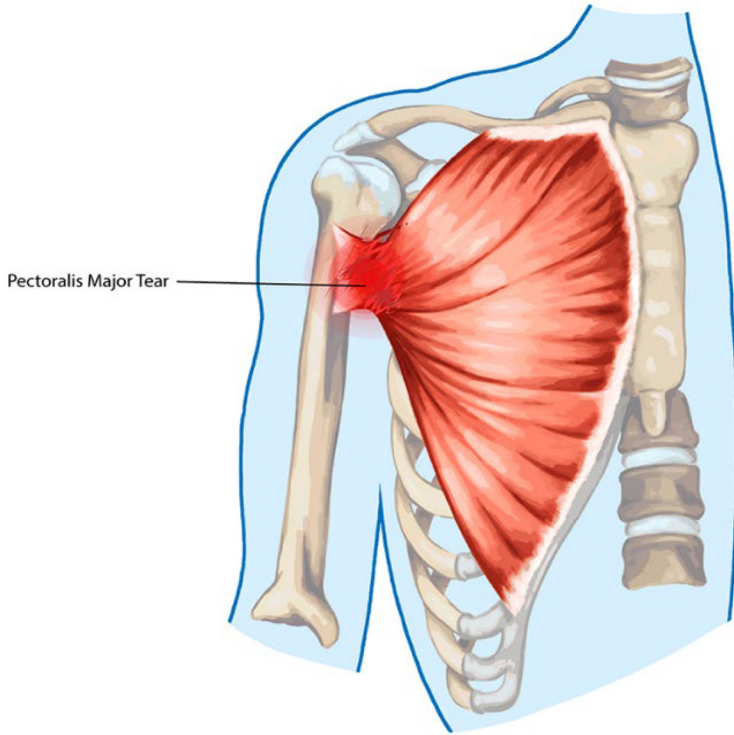
Common Shoulder Injuries in Bodybuilding



Injury Type	General Information	Symptoms	Relevant Risk Factors
AC Joint Injury ("Weightlifter's Shoulder")  Adapted from Pemporon/Shutterstock.com	Can occur insidiously or traumatically (falling with an adducted shoulder) that produces widening of the acromioclavicular (AC) joint, subchondral stress fractures, and potential bone lysis on the distal clavicle (collarbone) causing pain and functional disability, particularly with loading the upper extremity.	Anterior and/or superior shoulder pain near the AC joint; potential elevation of the distal clavicle if injury is severe enough; variable shoulder muscle weakness; may have shoulder impingement.	Benching >1.5x bodyweight, frequency (>1x per week), and duration of bench press (>5 years).
Anterior Shoulder Instability  Adapted from Pemporon/Shutterstock.com	Laxity of the ligamentous structures protecting the shoulder from anterior dislocation; as a result of laxity, partial subluxations or complete dislocations may occur with pain.	Apprehension/pain when the shoulder is placed in positions of abduction/external rotation or in the extended position.	Heavy bench pressing, dips, and any exercise requiring extension of the arm past the trunk, such as behind the neck pulldowns, places significant stress on the anterior shoulder.

Common Shoulder Injuries in Bodybuilding



Injury Type	General Information	Symptoms	Relevant Risk Factors
Pectoralis Major Tear 	A complete tearing (rupture) of the pectoralis major muscle tendon from the insertion on the humerus and occurs almost exclusively in males with or without use of anabolic steroids during bench press or high-velocity activities that place the shoulder in abduction/external rotation.	Inability to perform horizontal adduction (main movement of the pectoralis major); significant swelling and pain in the region.	Heavy bench pressing, dips and chest flies; anabolic steroid use.

Adapted from stihii/Wikimedia.org

All data from (Bak et al., 2000; Burkhart et al., 2003; Grossman et al., 2005; Kolber et al., 2010; König & Biener, 1990; Nevalainen et al., 2016)